

INDIAN INSTITUTE OF TECHNOLOGY HYDERABAD
Department of Mathematics

Final Exam

Date : October 19, 2024

MA1220: Calculus II

Time : 90 minutes

Max Marks : 40

Instructions:

- No specific charts, graphs, log book etc. required.
- Step marking depends on the procedure and the final answer. Hence, it is essential to show the detailed work.
- Answer all questions.

1. Find the area of the region in the first quadrant that is bounded above by $y = \sqrt{x}$ and below by the x -axis and the line $y = x - 2$. [3]

2. Show that if you cut a spherical loaf of bread into slices of equal width, each slice will have the same amount of (crust) outer surface area. [3]

3. Consider the improper integral [8]

$$\int_1^{\infty} \frac{\cos t}{t} dt.$$

Show that the improper integral is convergent. Discuss about the absolute convergence of this improper integral.

4. Consider the function [3]

$$f(x, y) = \begin{cases} \frac{5x^7y^8}{x^{21}+y^{12}} & \text{if } (x, y) \neq (0, 0), \\ 0 & \text{if } (x, y) = (0, 0). \end{cases}$$

Calculate $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$. Justify your answer.

5. Consider the function [8]

$$f(x, y) = \begin{cases} \frac{3x^2y}{x^2+y^2} & \text{if } (x, y) \neq (0, 0), \\ 0 & \text{if } (x, y) = (0, 0). \end{cases}$$

Show that the function is continuous at every point in $\mathbb{R}^2$, using the $\epsilon - \delta$ definition.

6. Show that at every point (x_0, y_0) in the domain of a differentiable function $f(x, y)$, the gradient of f is normal to the level curve through (x_0, y_0) . [3]

7. A flat circular plate has the shape of the region $x^2 + y^2 \leq 1$. The plate, including the boundary where $x^2 + y^2 = 1$, is heated so that the temperature at the point (x, y) is [6]

$$T(x, y) = 4x^3 - 15x^2 + 12x - 3y^2.$$

Find the temperatures at the hottest and coldest points on the plate.

8. Using Lagrange's multiplier method, find the rectangular box of the largest volume that has sides parallel to the x, y and z axes and that can be inscribed in the ellipsoid given by [6]

$$5x^2 + 3y^2 + 7z^2 = 1.$$